$$x(t) \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = x(t) * h(t)$$

$h(t)$

$$\sum_{k=0}^{M} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{N} b_k \frac{d^k x(t)}{dt^k}$$, order $= \max(M, N)$

$$y(t) = y_h(t) + y_p(t)$$

$$x(t) \longrightarrow \boxed{} \longrightarrow y(t)$$

$$x(t) = e^{st} \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau)\, d\tau$$

$h(t)$

$\sum_k a_k e^{s_k t}$

$\sum a_k H(s_k) e^{s_k t}$

$$= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)}\, d\tau$$

$$= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau) e^{-s\tau}\, d\tau}_{H(s)}$$

$$y(t) = H(s) e^{st}$$

eigenvalue eigenfunct^n

$Ax = \lambda x$

- Discrete Time Fourier Series (DTFS)

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n}$$

when $x[n]$ is discrete time N-periodic finite power signal

$$S_N \triangleq \left\{ x : \mathbb{Z} \longrightarrow \mathbb{C} \mid x[n+N] = x[n]\ \forall n,\ \frac{1}{N} \sum_{\langle N \rangle} |x[n]|^2 < \infty \right\}$$

To show: S_N is a V.S. (of dimens^n N) & $\{e^{jk\omega_0 n}\}$ are basis elements

To show: S_N is a V.S. (of dimens^n N) & $\{e^{jk\omega_0 n}\}$ are basis elements.

① S_N is a V.S.:

Pf: $(x+y)[n] \triangleq x[n]+y[n]$, $(\alpha x)[n] \triangleq \alpha x[n]$

i) Let $x, y \in S_N$

$(x+y)[n] = x[n]+y[n]$

$(x+y)[n+N] = x[n+N]+y[n+N] = x[n]+y[n] = (x+y)[n]$

ii) $$\frac{1}{N}\sum_{\langle N\rangle}|(x+y)[n]|^2 = \frac{1}{N}\sum_{\langle N\rangle}|x[n]+y[n]|^2$$
$$\leq \frac{1}{N}\sum_{\langle N\rangle}(|x[n]|+|y[n]|)^2$$
$$\leq \underbrace{\frac{1}{N}\sum_{\langle N\rangle}|x[n]|^2}_{<\infty}+\underbrace{\frac{1}{N}\sum_{\langle N\rangle}|y[n]|^2}_{<\infty}+\underbrace{\frac{1}{N}\sum_{\langle N\rangle}2|x[n]||y[n]|}_{<\infty}$$
$\because$ (A.M. $\geq$ G.M.)

$$\leq \infty$$

$\therefore$ i) & ii) $\Rightarrow x+y \in S_N$

② i) $(\alpha x)[n+N] = \alpha x[n+N] = \alpha x[n] = (\alpha x)[n]$

ii) $$\frac{1}{N}\sum_{\langle N\rangle}|\alpha x[n]|^2 = \underbrace{|\alpha|^2\frac{1}{N}\sum_{\langle N\rangle}|x[n]|^2}_{<\infty}$$

$\Rightarrow \alpha x \in S_N$

$\therefore$ S_N is a V.S. ✓

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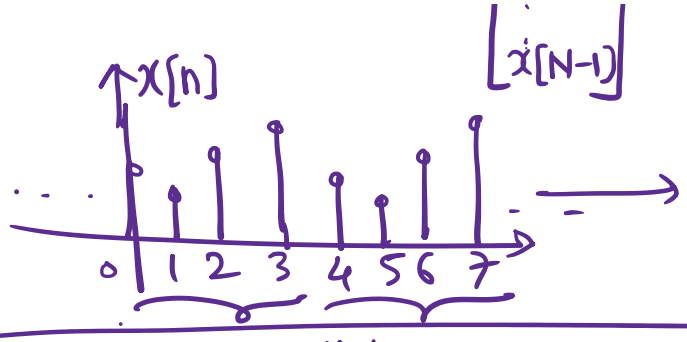
③ $\phi: S_N \longrightarrow \mathbb{C}^N \longrightarrow$ dimens$^n = N$

$x \longrightarrow \phi(x) \triangleq \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$

$\therefore$ dimens$^n(S_N) = N$

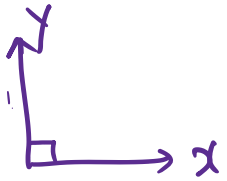
$\uparrow x[n]$



$$\longrightarrow \phi(x) = \begin{bmatrix} x[0] \\ \vdots \\ x[3] \end{bmatrix}$$

(III) $\{e^{jk\omega_0 n}\}_{k=0}^{N-1}$, $\omega_0 = \frac{2\pi}{N}$

$\underbrace{\text{Let } x_k[n] \triangleq e^{jk\omega_0 n}}_{\therefore x_k \in S_N}$, $\frac{1}{N}\sum_{\langle N\rangle} |x_k[n]|^2 = \frac{1}{N}\cdot N = 1$

Orthogonal $\Rightarrow$  $\Rightarrow$ L.I.

$\{x_k\}_{k=0}^{N-1}$, $\langle x, y\rangle \triangleq \frac{1}{N}\sum_{\langle N\rangle} x[n]\, y^*[n]$

$$\langle x_k, x_\ell\rangle = \frac{1}{N}\sum_{\langle N\rangle} x_k[n]\, x_\ell^*[n]$$

$$= \frac{1}{N}\sum_{\langle N\rangle} e^{jk\frac{2\pi}{N}n} e^{-j\ell\frac{2\pi}{N}n}$$

$$= \frac{1}{N}\sum_{\langle N\rangle} e^{j\underbrace{(k-\ell)}_{m}\frac{2\pi}{N}n} = \begin{cases} 1 & , k=\ell \\ 0 & , k\neq\ell \text{ (Roots of unity)} \end{cases}$$

$\Rightarrow \{x_k\}_{k=0}^{N-1}$ is orthogonal $\therefore$ L.I. set in S_N.

$\therefore \{x_k\}_{k=0}^{N-1}$ is a basis for S_N.

$$\Rightarrow x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n} \longrightarrow \text{D.T.F.S.}$$